Final Ethics Report and Responsible AI Usage Log

BIN381 Milestone 4 - SDHSA: Unemployment and Labour Market Exclusion

1. Purpose and scope

This report brings together the ethical issues we met in all four milestones, what we did about each, and what is still open. It covers the data (QLFS 2024 microdata), the descriptive analysis, the Power BI dashboard, the screening model, the Shiny app, and our own use of AI tools.

The principles we used to judge our choices are: respect for privacy, fairness between groups, honesty about what the results can and cannot show, transparency about how the model works, human oversight of any decision, and accountability for the work.

2. Ethical risk register

#	Phase	Risk	Who could be affected	What we did	Status
1	Business understanding	Describing unemployment by group could stigmatise those groups	Young people, rural residents, population groups with higher rates	Framed all findings as where support is needed, not as characteristics of people; neutral language; every group comparison described as an association	Mitigated
2	Data understanding	Re-identificatio n from small cells (for example rare age, province and group combinations)	Survey respondents	Data is a de-identified Stats SA extract; no names, addresses or ID numbers; small groups flagged in the dashboard (fewer than 1,000 records) and group sizes shown	Mitigated
3	Data understanding	Household hardship files (hunger, grants) are sensitive	Poor households	Not used (D07, D09 excluded)	Avoided
4	Preparation	Cleaning choices can change who is counted (dropping rows,	All groups	Every treatment documented with problem, treatment,	Mitigated

#	Phase	Risk	Who could be affected	What we did	Status
		recoding sentinels, imputation)		effect and verification; legitimate extreme values kept; 55 of 170,432 records removed and counted	
5	Preparation	Unweighted counts misstate population rates	All groups	All headline rates use survey weights; reproduced official rates to within 0.1 point	Mitigated
6	Preparation	Possible swap of formal/informal codes could lead to false statements about informal work	Informal workers	Field not used in any finding; flagged for checking against the codebook	**Open**
7	Modelling	Using gender and population group as model inputs	Groups protected by equality law	Kept so that fairness could be measured; outputs are advisory; model not to be used to decide eligibility	Accepted with safeguards
8	Modelling	Class imbalance (18.8% unemployed) makes accuracy misleading	People missed by the model	Judged on recall, precision, specificity, balanced accuracy and AUC; threshold analysis	Mitigated
9	Evaluation	Recall differs sharply by group (White 36%, Indian/Asian 37%, rural farms 50%, 35+ 64%, against 83% for Black African)	Groups the model finds less well	Gaps measured on six dimensions and disclosed in the report, model card, dashboard and Shiny app; other identification routes recommended	Disclosed, not removed
10	Evaluation	Reading associations as causes	Policy-makers, public	Non-causal wording throughout; recommendations written as priorities, not explanations	Mitigated

#	Phase	Risk	Who could be affected	What we did	Status
11	Deployment	Misuse of flags to decide who gets or loses support	Individuals	Banner on every page; threshold and “1 in 3 correct” shown; model card lists automated eligibility decisions as out of scope	Mitigated by design; depends on DEL practice
12	Deployment	Uploaded personal-level data stored or exposed	Respondents	App processes uploads in memory and does not save them; only monitoring counts are logged; hosting must be POPIA-compliant and access-controlled (no login in current version)	**Open for hosting**
13	Monitoring	Model decays or fairness gaps widen unnoticed	All groups	PSI, feature drift, missing-data, recall and group-gap checks; logged runs; retirement rule	Planned; needs an owner
14	Dashboard	Dashboard shows sample record counts; readers may quote them as national totals, and a selected province silently narrows every figure	Policy-makers, public	Sample and weighted figures reconciled in the final report (Table 4); the dashboard is described as a view of patterns and rankings; selected filters must be visible in every screenshot; weighted measures offered as an add-on	Mitigated; weighted measures recommended
15	Dashboard	Labels and text can overstate (a youth-only trend read as national; “Total Employees” used for all records; an employment-ty	Policy-makers	Wording corrected in the report; table flagged as unverified; quarterly chart to be sorted by quarter	**Open** until the dashboard is relabelled

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		pe table built on the unverified formal/informal field)			
16	Our own AI use	Plausible but wrong output, unattributed text or code	The team, the lecturer	Log below; team checks and runs all outputs; AI-computed figures reproduced by R scripts	In progress (see section 5)

3. Privacy and POPIA

- **Data source and consent.** The QLFS is collected by Statistics South Africa under the Statistics Act (No. 6 of 1999). We used a de-identified academic extract. HouseholdID is a survey-internal identifier that cannot be traced to a person without Stats SA's master file.
- **Data minimisation.** Only four of eleven files were used, and the final dashboard draws on the two cleaned analysis tables. Identifier and weight columns are removed before modelling.
- **Small groups.** Groups with fewer than 1,000 records carry a warning in the dashboard.
- **The app.** Uploaded files are scored in the running R session and are not written to disk. The monitoring log stores only counts, PSI and recall.
- **If the app is hosted.** Before real person-level data is loaded, the host must (a) have a lawful basis and purpose under POPIA, (b) restrict access with login, (c) use encrypted transport, and (d) have a retention rule for any logs. These are not built into the current version.

4. Fairness: what we found and decided

Dimension	Recall gap	Our decision
Gender	0.2 points	No action needed
Age (youth vs 35+)	26 points	Accepted: matches the business priority (youth) but 35+ are under-detected; reviewed by hand
Province	22 points (66% to 87%)	Disclose; consider provincial review of flags
Population group	46 points (36% to 83%)	Disclose; do not rely on the model for the lowest-recall groups; use other identification routes
Area type	32 points (farms 50% vs rural traditional 82%)	Disclose; treat farm-area flags as lower confidence
Quarter	4 points	No action needed

Why we did not set different thresholds by group. It would raise recall for some groups, but it means treating people differently by race in a screening tool. That decision belongs to the department with legal advice, not to a student project. We recommend DEL decides it at programme level.

Why we kept population group as an input. Removing it would not remove the pattern, because other fields (province, area type, education) carry related information, and we would lose the ability to measure the gap.

5. Responsible AI usage log

We used an AI assistant for editing,formating and error checking. The table lists what it was used for. “Team check” says how the team confirms the output. A box is ticked only where the check was actually done, and the entry says what was compared and when. Boxes that need R to be run are left open and name who runs them. Tick one only after you have really done the check; if you cannot, change the entry to say so. Each member should confirm the rows that cover their own work.

Stage	What the AI did	What the team did	Team check	Done
M1 Data understanding	Code structure suggestions	Ran all R code; inspected data	Outputs reproduced in R	Done
M2 Data preparation	Pipeline scaffolding, debugging	Made cleaning decisions; verified	Verification chunks in the notebook	Done
M3 Modelling	Model configuration, threshold analysis	Trained and tuned in R	Results re-run; seeds fixed	Done
M3 Evaluation	Metric selection, interpretation help	Chose final model	Metrics checked against R output	Done
M4 Dashboard	used to review the dashboard text and numbers against the cleaned data	built the .pbix and wrote the findings and design sections	Dashboard figures recomputed from the cleaned panel (sample rates 32.9%, quarterly 33.1, 33.8, 32.4, 32.2; youth NEET 42.1%) and reconciled with weighted figures	Done
M4 Data, modelling and Shiny sections (Member 2)	used to cross-check data and data modelling with the Milestone 3 outputs	wrote the data, modelling, deployment and limits text and took the screenshots	Row counts, model table and fairness tables compared with Milestone 3 outputs	Done
M4 Final report	structured the integrated text	Edited the report	Read every number against the notebook and outputs	Done

Statement of responsibility. All AI-assisted outputs must be reviewed, tested and adapted by the team. The team takes responsibility for the correctness, originality and appropriateness of the content, code and conclusions. AI tools were not used to hide limitations; several limitations in this project (the failed join, the accuracy trap, the fairness gaps, the formal/informal question) were stated openly in the work.

6. Model card summary (for the Shiny app and report)

- **Model:** random forest (ranger, 500 trees) trained on QLFS 2024 person-quarter records, working-age (15 to 64), n = 170,377 (test set 51,114).
- **Intended use:** decision support for outreach and prioritisation; analysts review every flag.
- **Out of scope:** automatic decisions about benefits or eligibility; use outside South African labour-market data; use on data from a different survey design.
- **Operating threshold:** 0.20 (recall 80.0%, precision 33.8%, specificity 63.6%, ROC-AUC 0.787).

- **Known weaknesses:** lower recall for White, Indian/Asian, farm-area and 35+ groups; about 45% of people flagged; trained on one year.
- **Monitoring and retirement:** see report section 5.